ARDOR / INDEPENDENT FUNCTION MODULE

Mono line output / relay mute / internal amp feed / HAND ASSEMBLY

T1 | HAND ASSEMBLY | SEE PIN GUIDE AT RIGHT

PIN GUIDE: IN = into this board / OUT = out / I/O = both / GND = 0 V

J101 / PWR + ENABLE IN
1: 5V IN 2: GND 0V
3: ENABLE IN

J102 / AUDIO IN
1: AUDIO IN 2: GND 0V

J503 / LINE OUT
1: LINE OUT 2: GND 0V

J502 / AMP AUDIO OUT
1: AUDIO OUT 2: GND 0V

J102
MONO AUDIO INPUT FROM SOURCE

C401
10u / 50V BIPOLAR

U501A
MCP6022-I/P

C501
47u / 25V polarized

R502
100R / 1% 0.25W

K501
Omron G5V-1 DC5

J503
LINE AUDIO OUTPUT TO TS SOCKET

R401
100kR / 1% 0.25W

VREF

R501
10kR / 1% 0.25W

CHASSIS
D501
1.5KE6.8CA

Q501
2N3904 / E-B-C

R503
470R / 1% 0.25W

D502
1N4148

R504
100kR / 1% 0.25W

J101
5V POWER + ENABLE INPUT

R405
10kR / 1% 0.25W

U501B
MCP6022-I/P

R406
10kR / 1% 0.25W

C403
10u / 25V polarized

FB101
800R@100MHz / WE 7427501

C101
22u / 25V polarized

C102
100n / 50V X7R

U501C
MCP6022-I/P

R101
0R / 1% 0.25W

C404
100n / 50V X7R

J502
AUDIO OUTPUT TO AMP INPUT

C502
10u / 50V BIPOLAR

R505
1kR / 1% 0.25W

R506
100kR / 1% 0.25W

Input <=1 Vrms, input impedance ~100k. LINE load >=10k; internal AMP load >=100k.
Relay OFF: jack tip grounded, source disconnected. Relay ON: unity mono output. No true bypass.
3.3 V active-high enable; leave LOW until audio is stable. 5 V supply budget >=50 mA.
C501 + toward buffer. J502 is a signal feed, not a speaker driver; jack sleeves bond the enclosure.

Engineering prototype — bench validation required

Ardor

Sheet: /
File: Ardor_Line_Out_THT.kicad_sch

Title: Mono line output / relay mute / internal amp feed / HAND ASSEMBLY

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